

Basic Electronics Engineering

(Course Code: ECL101)

Lecture 10-12: MOSFET

Course Instructor:

Vivek Raghuwanshi, Ph.D.

Email: vivekr@iitbhilai.ac.in

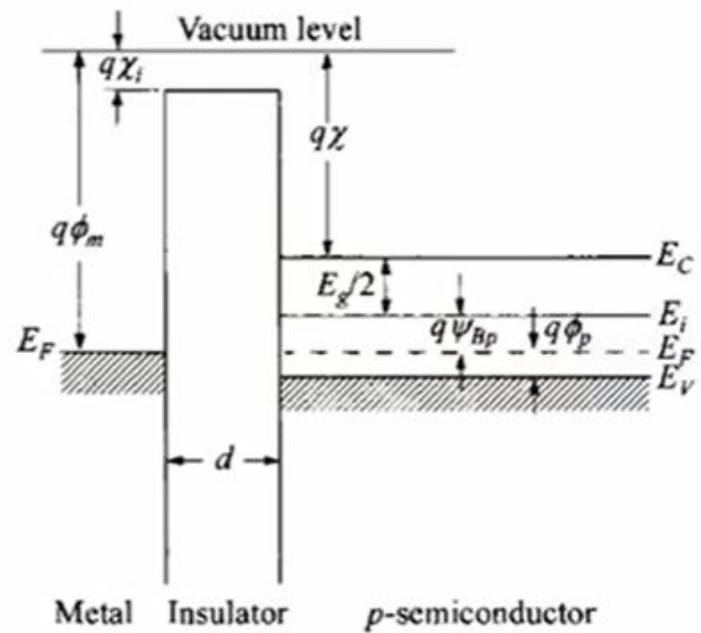
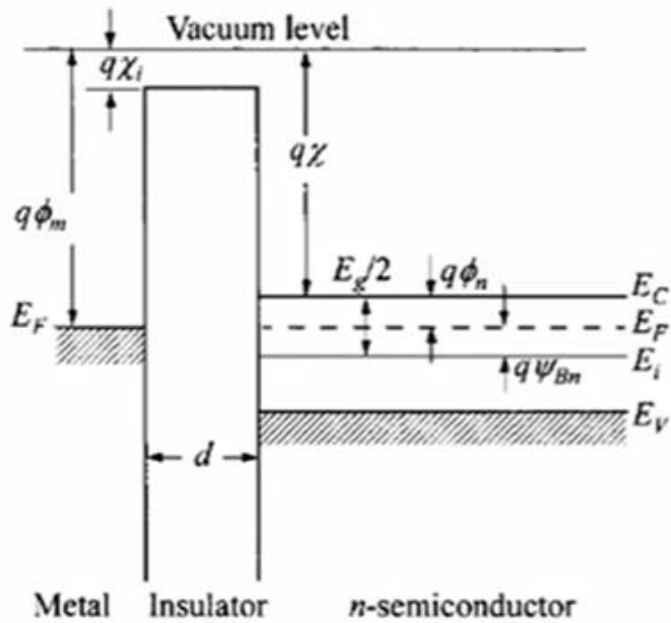
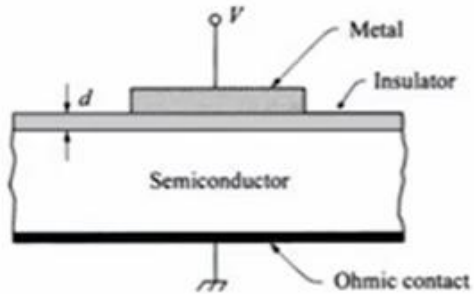
Office: ED1, 602

Webpage: <https://www.iitbhilai.ac.in/index.php?pid=vivekr>

Note: The information provided in the slides are taken mainly from text books (R. Boylested, Sedra Smith... and Millman-Halkias...) other resources from internet, for teaching/academic use only

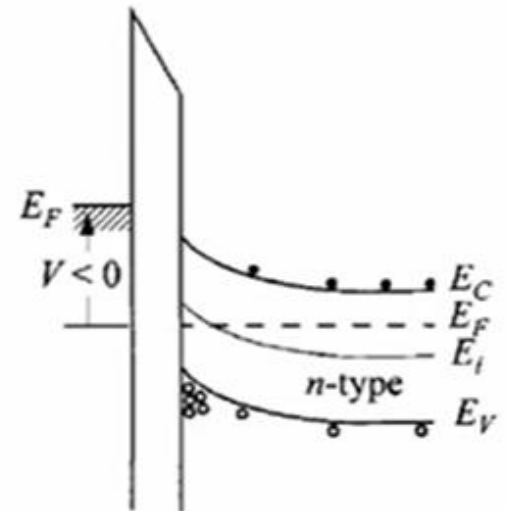
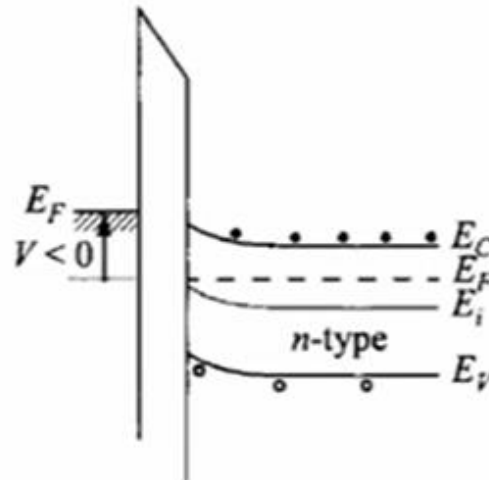
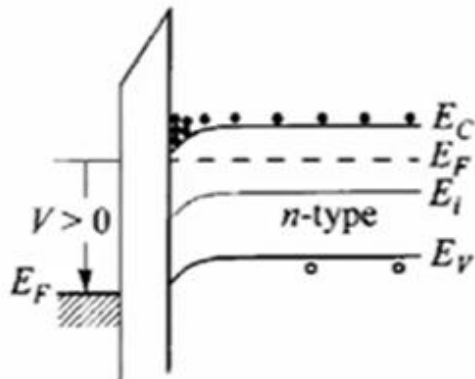
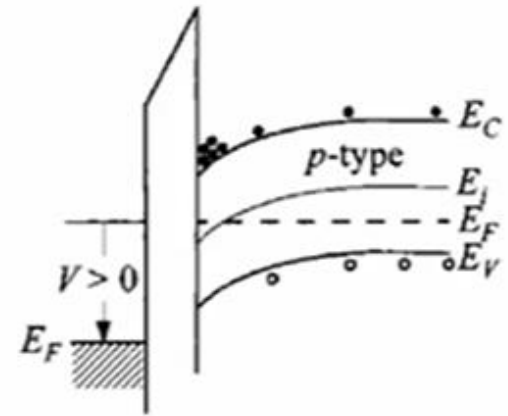
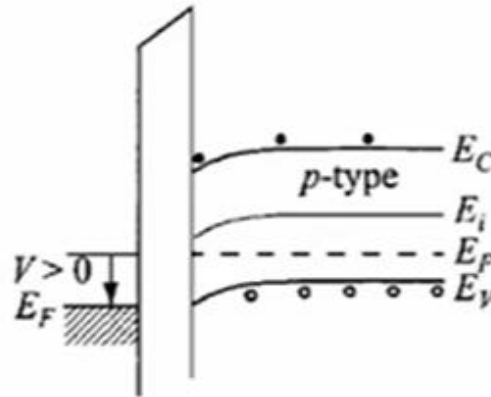
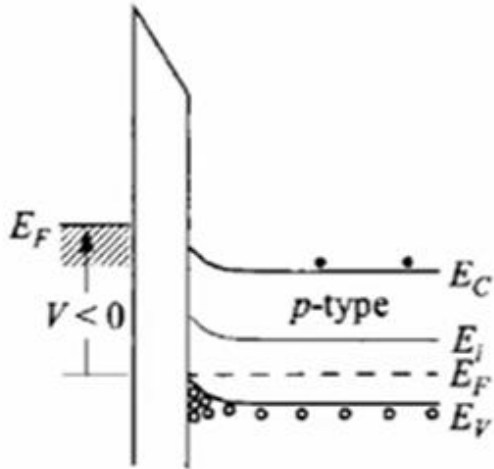
Metal Oxide Semiconductor Field-Effect Transistors (MOSFETs)

MOS Capacitor



Regions of operation

1. Accumulation
2. Depletion
3. Saturation

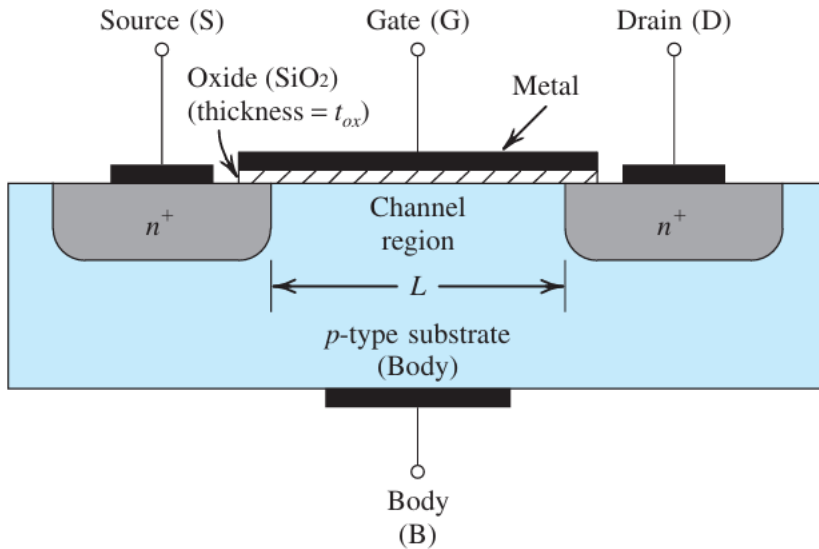
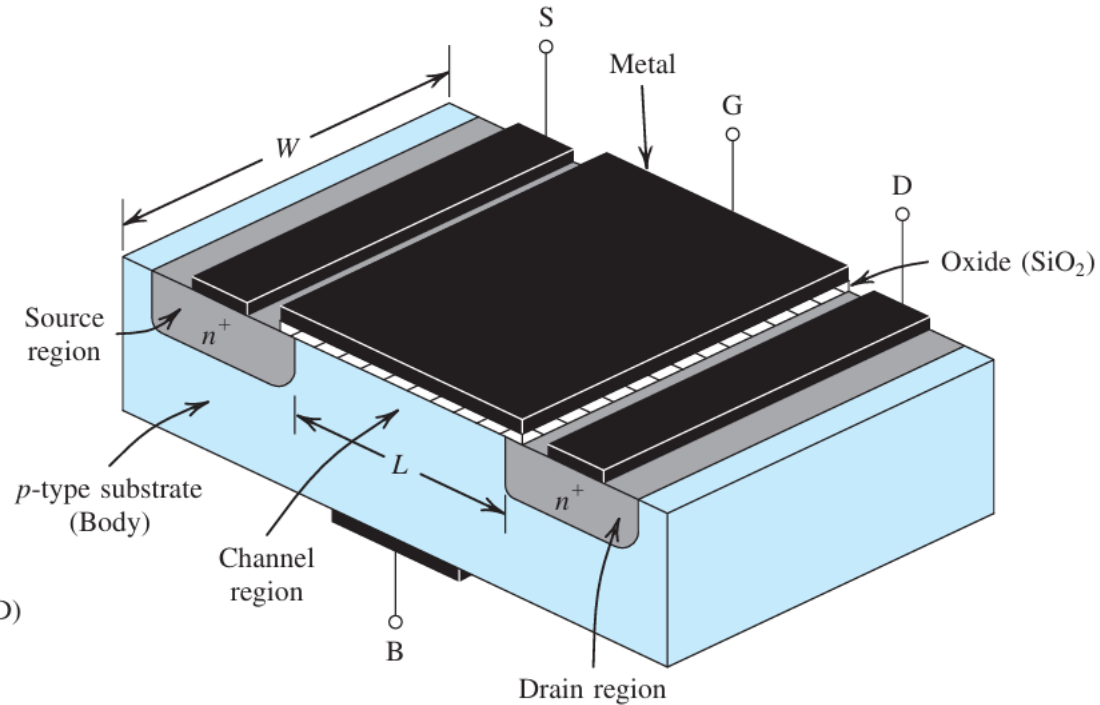


MOSFET

Device Structure

Four Terminals:

- 1) Gate (G)
- 2) Source (S),
- 3) Drain (D),
- 4) Substrate or body terminal (B)

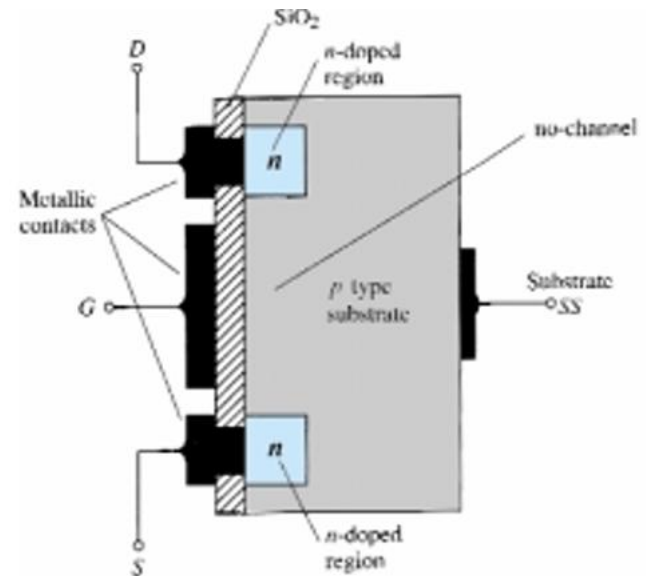
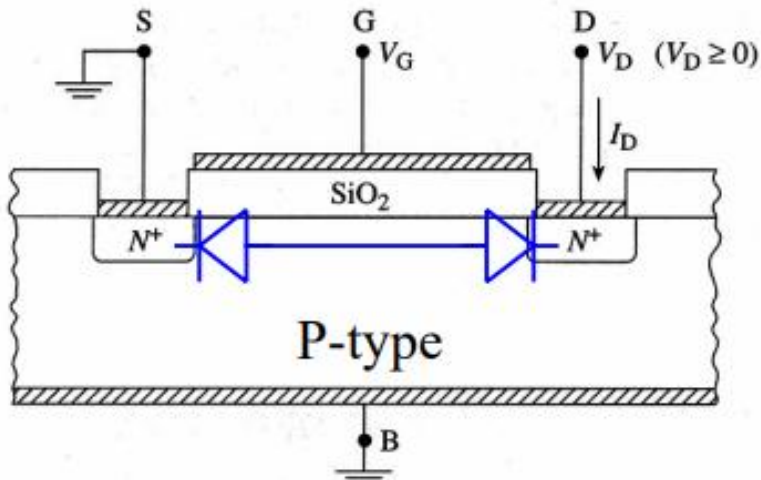


Thickness of *t_{ox}* (typically 1nm to 10nm)

n-channel enhancement-type MOSFET

Operation with Zero Gate Voltage

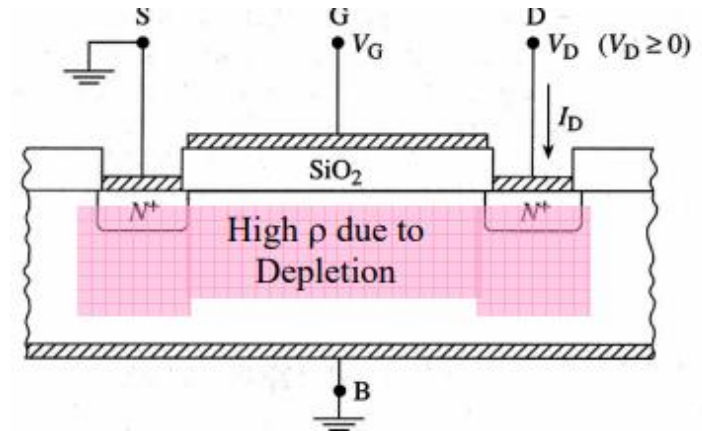
- With $V_{GS} \leq 0$, two back-to-back diodes exist in series between drain and source.
- One diode is formed by the pn junction between the n+ drain region and the p-type substrate, and the other diode is formed by the pn junction between the p-type substrate and the n+ source region
- These back-to-back diodes prevent current conduction from drain to source when a voltage v_{DS} is applied



Channel Formation

When $0 < V_{GS} \leq V_t$

(depletion), there is a deficit of electrons and holes making the channel very highly resistive. $\Rightarrow$ No Drain current can flow

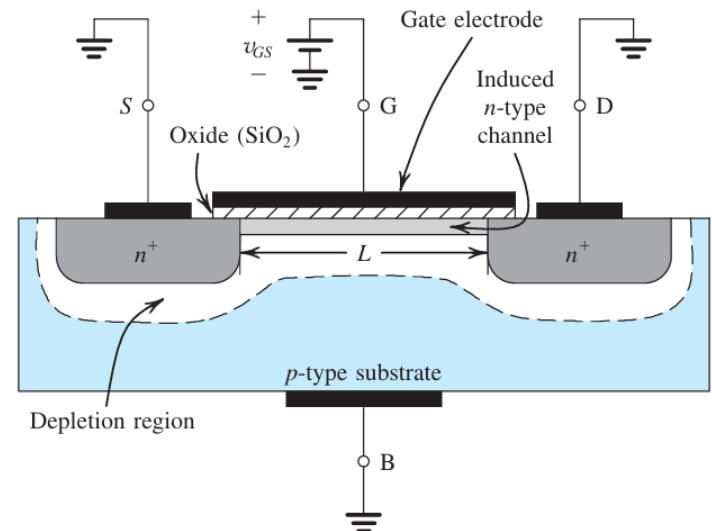


The value of V_{GS} at which a sufficient number of mobile electrons accumulate in the channel region to form a conducting channel is called the threshold voltage (V_t)

V_t for an n-channel FET is positive.

If both V_{DS} and V_{GS} is set at some positive voltage greater than 0 V, then the positive potential at the gate will pressure the holes (since like charges repel) in the *p*-substrate along the edge of the SiO₂ layer to leave the area and enter deeper regions of the *p*-substrate.

However, the electrons in the *p*-substrate (the minority carriers of the material) will be attracted to the positive gate and accumulate in the region near the surface of the SiO₂ layer. The SiO₂ layer and its insulating qualities will prevent the negative carriers from being absorbed at the gate terminal.

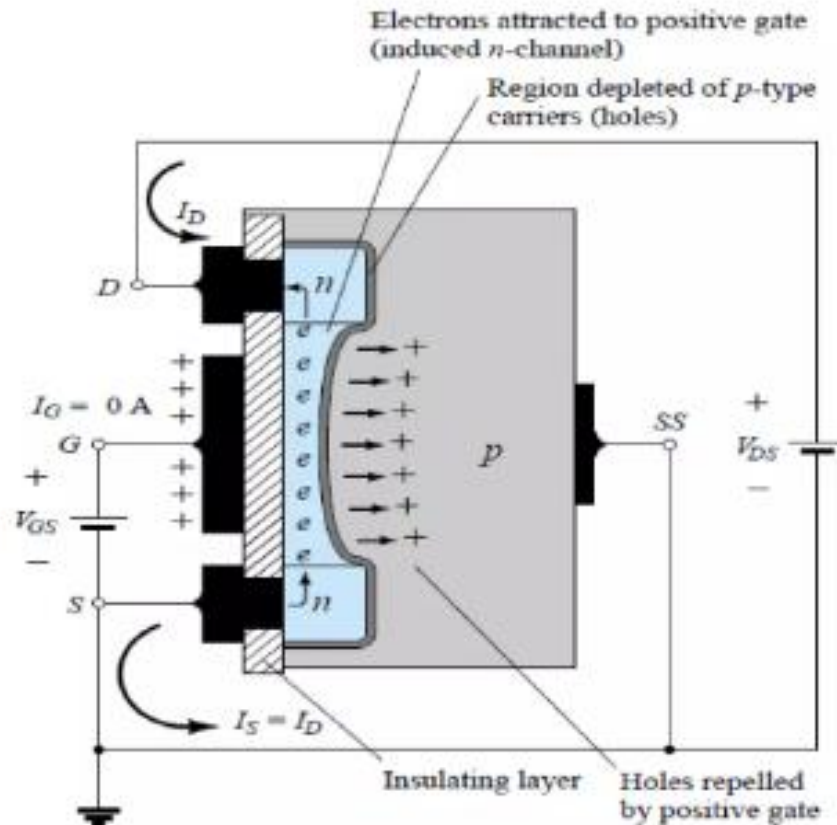


Channel Formation

When $V_{GS} \geq V_t$ (Inversion)

As V_{GS} increases in magnitude, the concentration of electrons near the SiO_2 surface increases until eventually the induced n-type region can support a measurable flow between drain and source. The level of V_{GS} that results in the significant increase in drain current is called the threshold voltage and is given the symbol V_T .

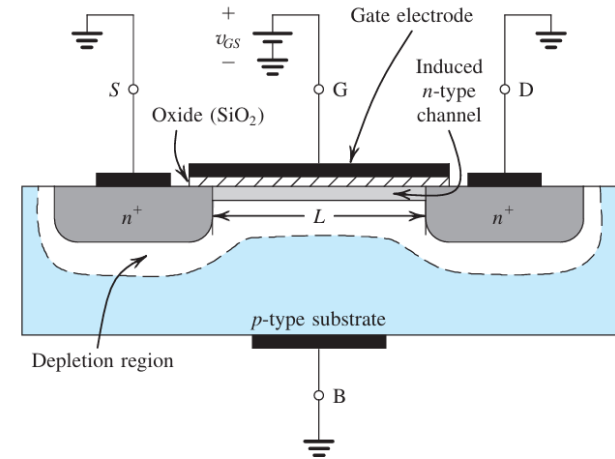
Since the channel is nonexistent with $V_{GS}=0$ V and “enhanced” by the application of a positive gate-to-source voltage, this type of MOSFET is called an enhancement-type MOSFET.



The voltage across this parallel-plate capacitor, that is, the voltage across the oxide, must exceed V_t for a channel to form

- When $v_{DS} = 0$, as in Fig, the voltage at every point along the channel is zero, and the voltage across the oxide (i.e., between the gate and the points along the channel) is uniform and equal to v_{GS}
- The excess of v_{GS} over V_t is termed the effective voltage or the **overdrive voltage** and is the quantity that determines the charge in the channel. we are denoting $(v_{GS} - V_t)$ by V_{ov}

$$v_{GS} - V_t \equiv v_{OV}$$



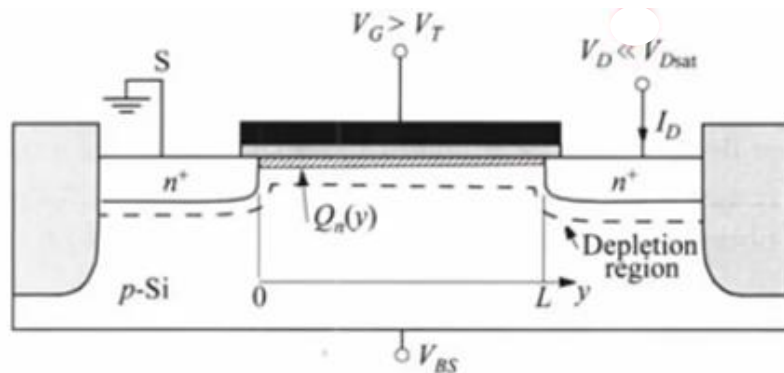
We can express the magnitude of the electron charge in the channel by

$$|Q| = C_{ox}(WL)v_{OV}$$

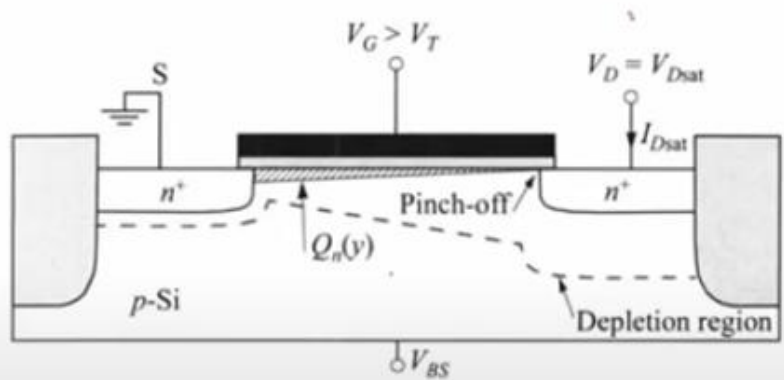
$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

where C_{ox} , called the oxide capacitance, is the capacitance of the parallel-plate capacitor per unit gate area

Effect of Increasing V_{DS} at fixed V_{GS}

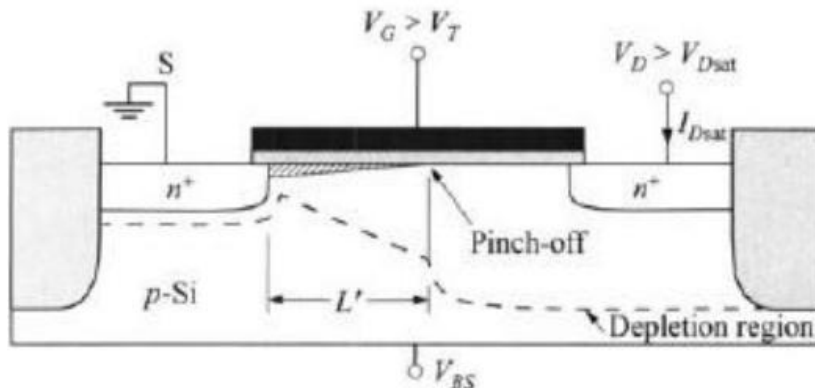


Effect: The channel acts as a resistor, $I_d \propto V_d$. This is the linear region.



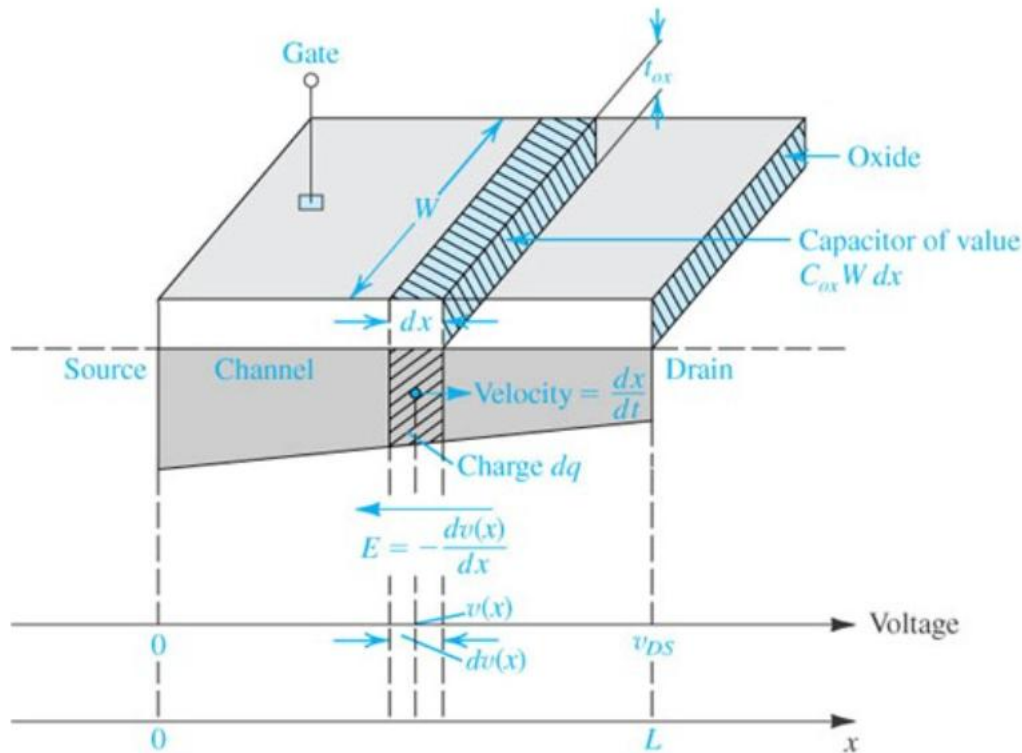
Effect: The inversion charge at the drain end $Q_n(L) = 0$

Pinch off point occurs there because the relative voltage between the gate and the semiconductor is reduced



Effect: the pinch-off point starts to move toward the source, but the pinch-off point voltage is the same

Derivation of Drain Current Equation



$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

$$dq = -C_{ox}(Wdx)[v_{GS} - v(x) - V_t]$$

$$E(x) = -\frac{dv(x)}{dx}$$

$$\frac{dx}{dt} = -\mu_n E(x) = \mu_n \frac{dv(x)}{dx}$$

$$i = \frac{dq}{dt} = \frac{dq}{dx} \frac{dx}{dt}$$

$$i = -\mu_n C_{ox} W \left[v_{GS} - v(x) - V_t \right] \frac{dv(x)}{dx}$$

$$i_D = -i = \mu_n C_{ox} W \left[v_{GS} - v(x) - V_t \right] \frac{dv(x)}{dx}$$

$$i_D dx = \mu_n C_{ox} W \left[v_{GS} - V_t - v(x) \right] dv(x)$$

$$\int_0^L i_D dx = \int_0^{v_{DS}} \mu_n C_{ox} W \left[v_{GS} - V_t - v(x) \right] dv(x)$$

$$i_D = (\mu_n C_{ox}) \left(\frac{W}{L} \right) \left[(v_{GS} - V_t) v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$

$$dq = -C_{ox}(Wdx) \left[v_{GS} - v(x) - V_t \right]$$

$$E(x) = -\frac{dv(x)}{dx}$$

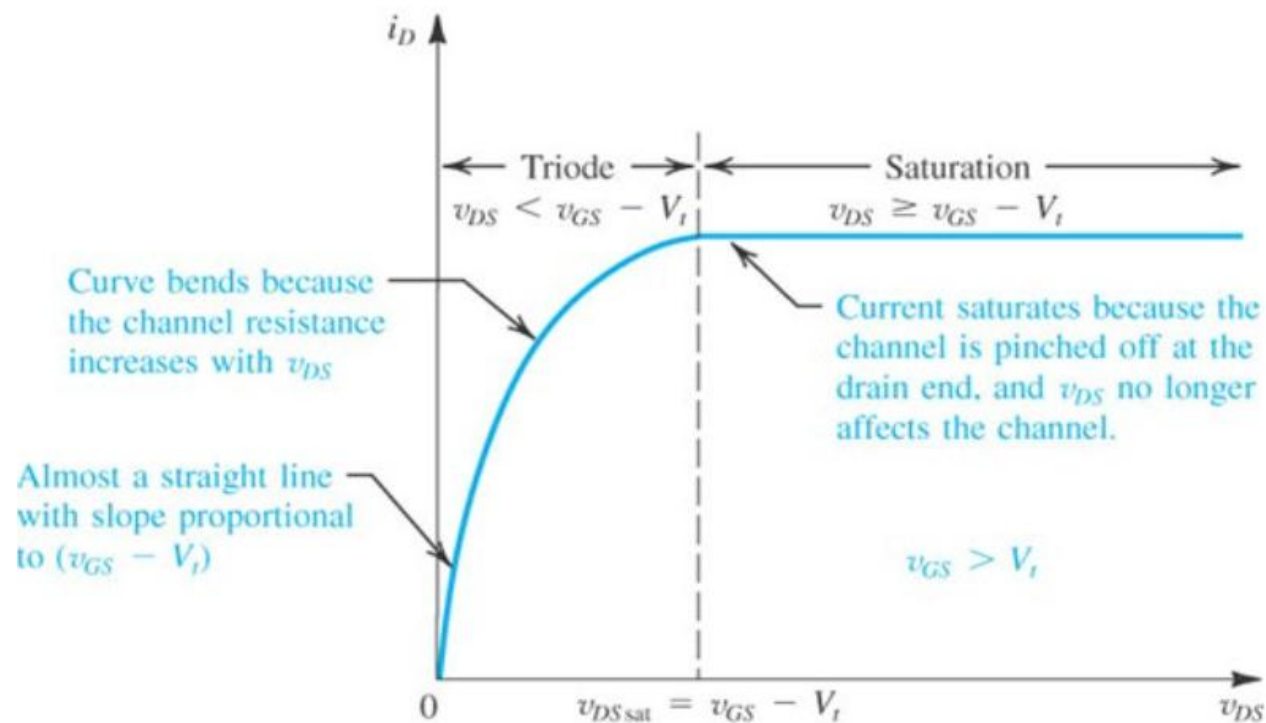
$$\frac{dx}{dt} = -\mu_n E(x) = \mu_n \frac{dv(x)}{dx}$$

process transconductance parameter k'_n

$$k'_n = \mu_n C_{ox}$$

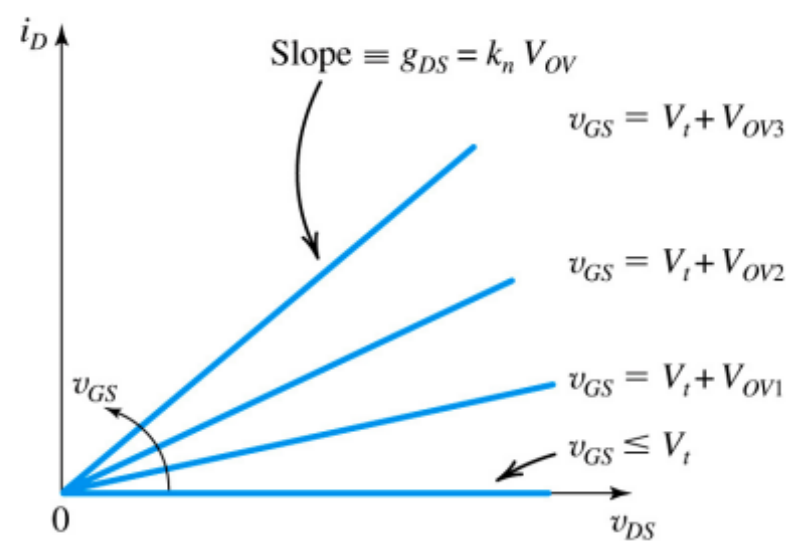
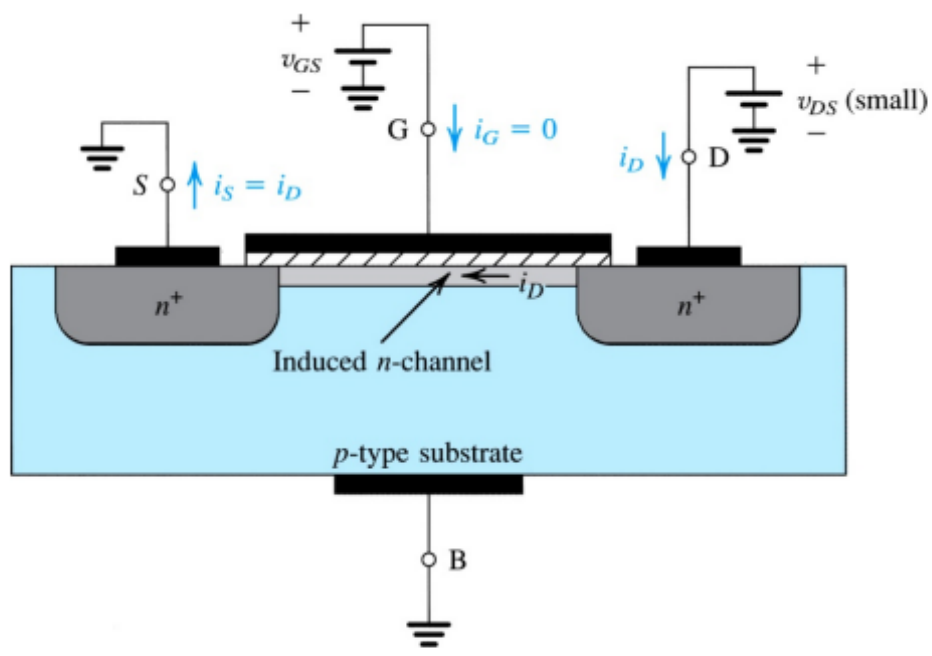
$$i_D = k'_n \left(\frac{W}{L} \right) \left[(v_{GS} - V_t) v_{DS} - \frac{1}{2} v_{DS}^2 \right] \quad (\text{Triode region})$$

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (v_{GS} - V_t)^2 \quad (\text{saturation region})$$



$$v_{DSsat} = v_{GS} - V_t$$

For a Small V_{DS}



- When a small positive bias is applied to a drain:

channel conductance is proportional to the overdrive voltage ($V_{OV}=V_{GS}-V_{th}$)
and drain-source voltage (V_{DS})

$$i_D = \left[(\mu_n C_{ox}) \left(\frac{W}{L} \right) (v_{GS} - V_t) \right] v_{DS}$$

The conductance g_{DS} of the channel

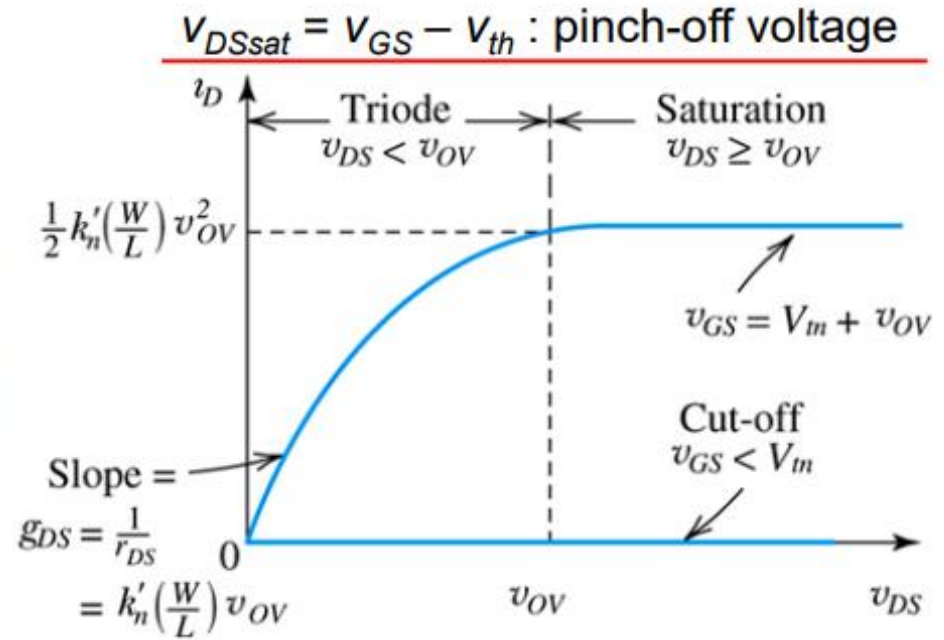
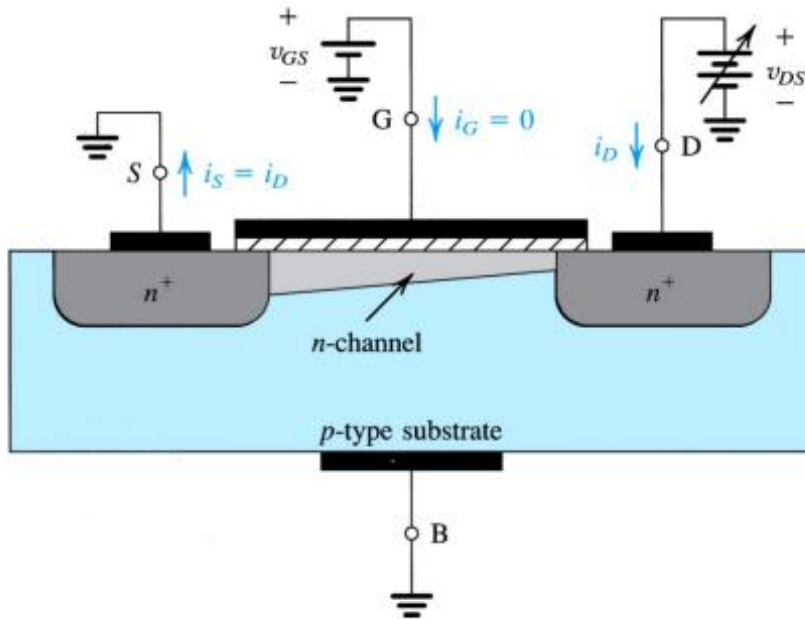
$$g_{DS} = (\mu_n C_{ox}) \left(\frac{W}{L} \right) (v_{GS} - V_t)$$

MOSFET transconductance parameter k_n

$$k_n = k'_n (W/L)$$

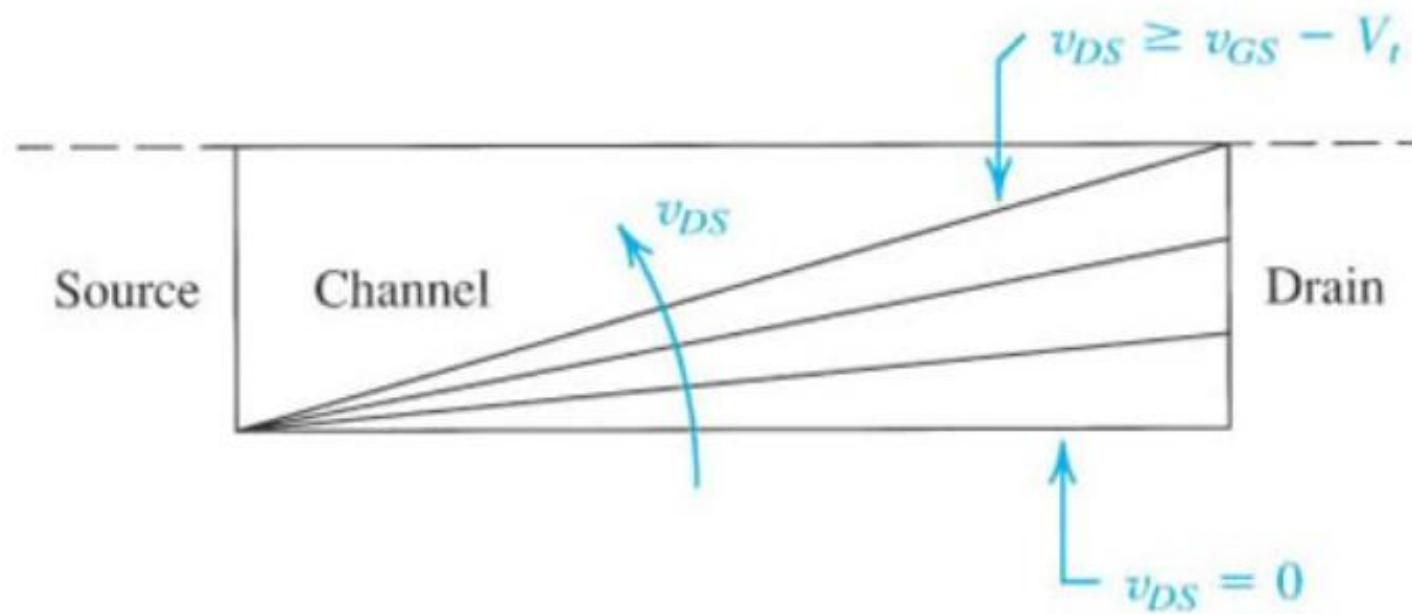
$$k_n = (\mu_n C_{ox}) (W/L)$$

For a large V_{DS}



- When a large positive bias ($> V_{GS} - V_{th}$) is applied to a drain:

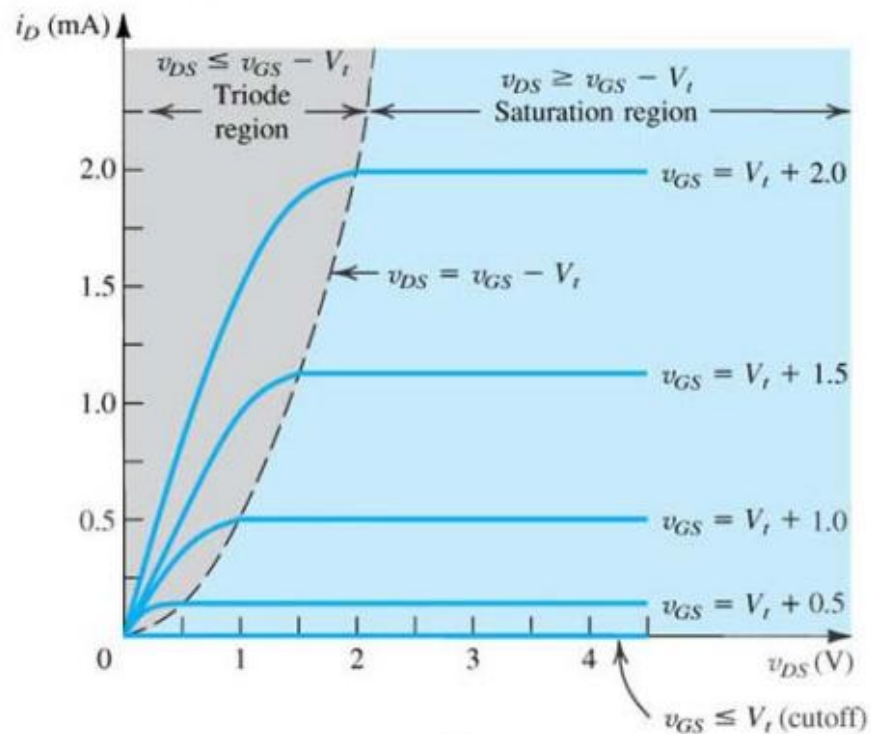
channel is pinched off. Increasing V_{DS} beyond the pinch-off voltage has little effect on the channel shape.



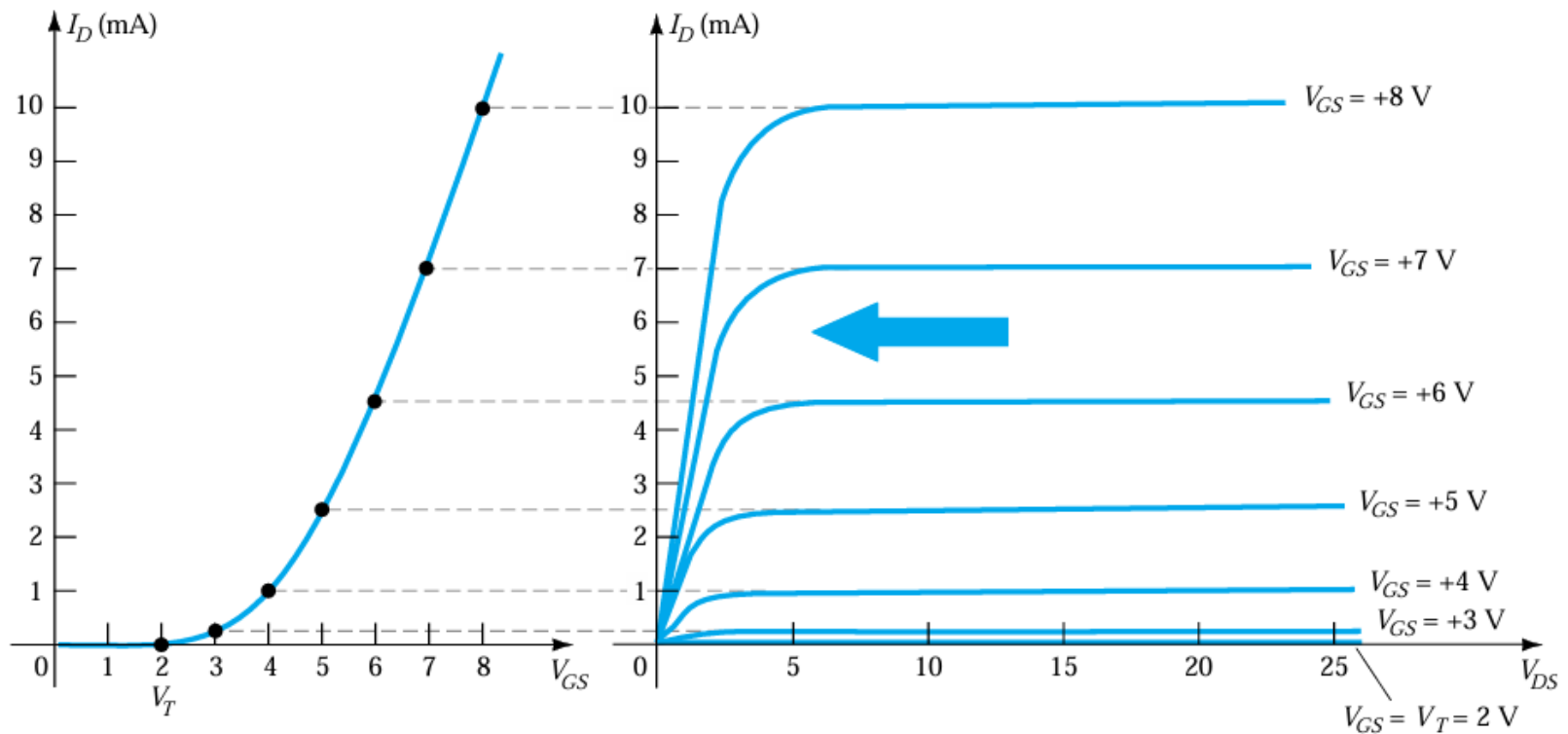
$$i_D = k'_n \left(\frac{W}{L} \right) \left[(v_{GS} - V_t) v_{DS} - \frac{1}{2} v_{DS}^2 \right] \quad \begin{matrix} V_{GS} > V_{th}, & V_{DS} < V_{GS} - V_{th} \\ \text{(Triode region)} \end{matrix}$$

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (v_{GS} - V_t)^2 \quad \begin{matrix} V_{GS} > V_{th}, & V_{DS} > V_{GS} - V_{th} \\ \text{(saturation region)} \end{matrix}$$

$$k'_n = \mu_n C_{ox}$$



I-V Characteristics of N channel MOSFET



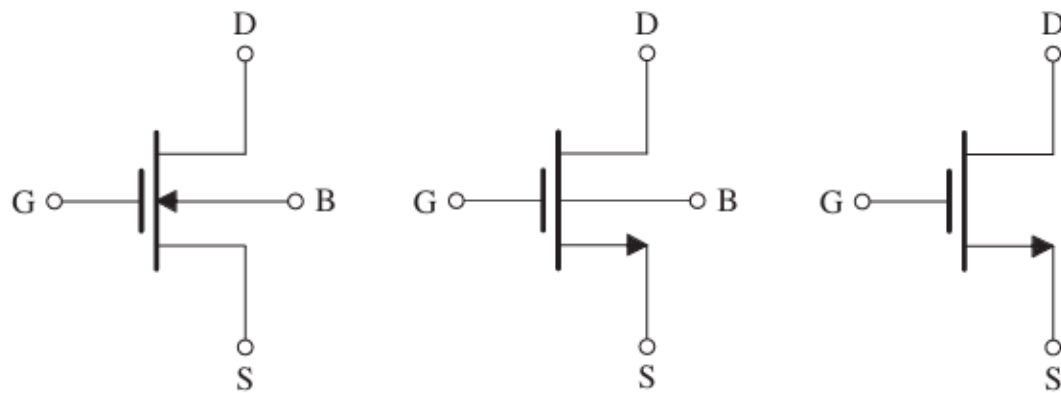
V_{GS} is always positive

As V_{GS} increases, I_D increases

But if V_{GS} is kept constant and V_{DS} is increased, then I_D saturates

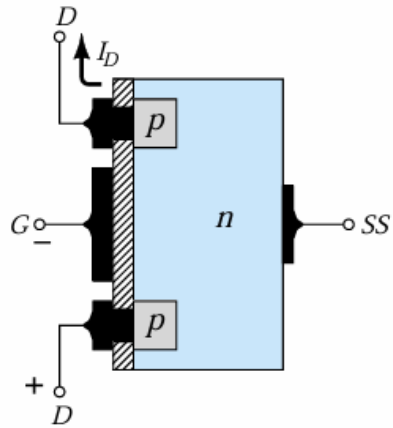
The saturation level, V_{DSsat} is reached.

$$V_{DSsat} = V_{GS} - V_T$$

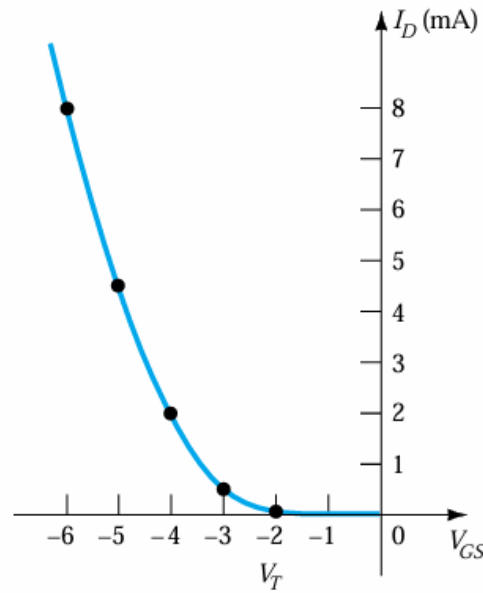


Symbols for the n-channel enhancement-type MOSFET

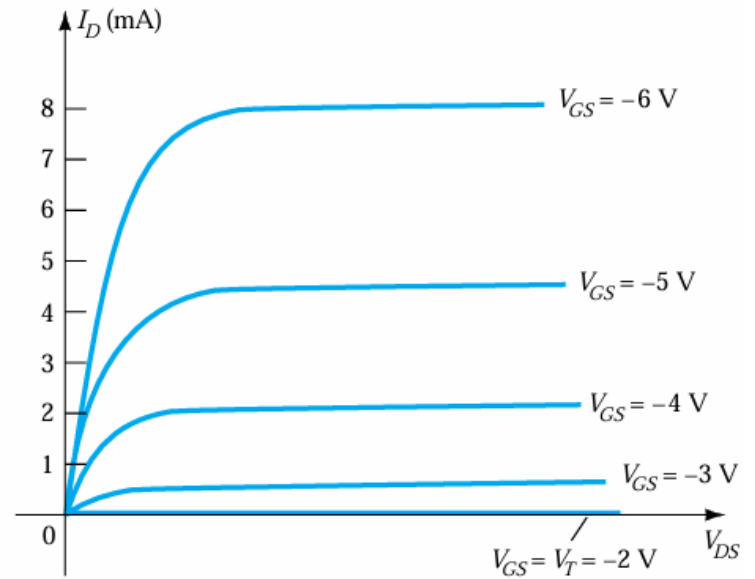
P channel MOSFET



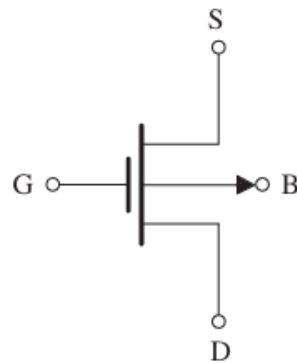
(a)



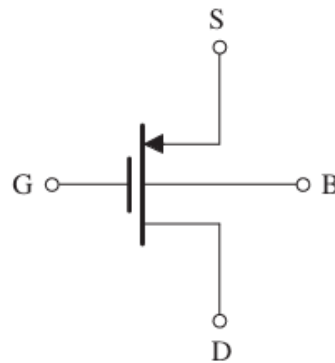
(b)



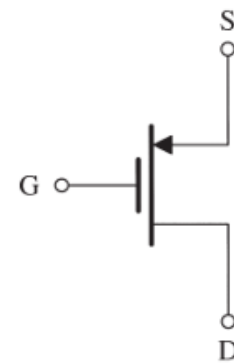
(c)



(a)



(b)



(c)

Consider a process technology for which $L_{\min} = 0.4 \mu\text{m}$, $t_{ox} = 8 \text{ nm}$, $\mu_n = 450 \text{ cm}^2/\text{V} \cdot \text{s}$, and $V_t = 0.7 \text{ V}$.

- (a) Find C_{ox} and k'_n .
- (b) For a MOSFET with $W/L = 8 \mu\text{m}/0.8 \mu\text{m}$, calculate the values of V_{OV} , V_{GS} , and $V_{DS\min}$ needed to operate the transistor in the saturation region with a dc current $I_D = 100 \mu\text{A}$.
- (c) For the device in (b), find the values of V_{OV} and V_{GS} required to cause the device to operate as a $1000\text{-}\Omega$ resistor for very small v_{DS} .

Solution

(a)

$$\begin{aligned}C_{ox} &= \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.45 \times 10^{-11}}{8 \times 10^{-9}} = 4.32 \times 10^{-3} \text{ F/m}^2 \\&= 4.32 \text{ fF}/\mu\text{m}^2 \\k'_n &= \mu_n C_{ox} = 450 (\text{cm}^2/\text{V} \cdot \text{s}) \times 4.32 (\text{fF}/\mu\text{m}^2) \\&= 450 \times 10^8 (\mu\text{m}^2/\text{V} \cdot \text{s}) \times 4.32 \times 10^{-15} (\text{F}/\mu\text{m}^2) \\&= 194 \times 10^{-6} (\text{F}/\text{V} \cdot \text{s}) \\&= 194 \mu\text{A}/\text{V}^2\end{aligned}$$

(b) For operation in the saturation region,

$$i_D = \frac{1}{2} k'_n \frac{W}{L} V_{OV}^2$$

Thus,

$$100 = \frac{1}{2} \times 194 \times \frac{8}{0.8} V_{OV}^2$$

which results in

$$V_{OV} = 0.32 \text{ V}$$

Thus,

$$V_{GS} = V_t + V_{OV} = 1.02 \text{ V}$$

and

$$V_{DSmin} = V_{OV} = 0.32 \text{ V}$$

(c) For the MOSFET in the triode region with v_{DS} very small,

$$r_{DS} = \frac{1}{k'_n \frac{W}{L} V_{OV}}$$

Thus

$$1000 = \frac{1}{194 \times 10^{-6} \times 10 \times V_{OV}}$$

which yields

$$V_{OV} = 0.52 \text{ V}$$

Thus,

$$V_{GS} = 1.22 \text{ V}$$